Test AI: Algoritmi si Tehnici

Subiecte propuse:

1. Categorie: N-QUEENS

Intrebare: For the 4-Queens problem with partial assignment [(4, 1), (3, 2)], which strategy should be used to complete the solution?

2. Categorie: GENERALISED_HANOI

Intrebare: For the Towers of Hanoi with 9 disks and 3 pegs, which strategy minimizes the number of moves?

3. Categorie: GRAPH_COLORING

Intrebare: Given a graph with 10 nodes and 4 colors, how would you assign colors to avoid conflicts?

4. Categorie: KNIGHTS_TOUR

Intrebare: For a 7x7 chessboard, which strategy is most suitable to generate a Knight's Tour?

5. Categorie: CSP_BACKTRACKING

Intrebare: Given variables ['X1', 'X2', 'X3', 'X4', 'X5'], domains {'X1': [1, 2, 3], 'X2': [1, 2, 3], 'X3': [1, 2, 3], 'X4': [1, 2, 3], 'X5': [1, 2, 3]}, constraints ['X1!=X3', 'X1!=X4', 'X1!=X5', 'X

6. Categorie: MINMAX_ALPHABETA

Intrebare: For a MinMax tree of depth 3 with leaf values [-1, 4, -2, -7, 1, -3, -5, 8], what would be the root value using Alpha-Beta pruning?

7. Categorie: GAME_THEORY

Intrebare: Given the game matrix [[(1, 2), (2, 5), (4, 1)], [(5, 5), (5, 0), (4, 3)], [(4, 0), (5, 3), (1, 3)]], is there a pure Nash equilibrium, and if so, which one?

Grila de Corectare:

Raspuns 1 (n-queens):

Backtracking with Forward Checking (FC) and Minimum Remaining Values (MRV)

Raspuns 2 (generalised_hanoi):

Recursive optimal 3-peg algorithm

Raspuns 3 (graph_coloring):

DSATUR heuristic (Degree of Saturation)

Raspuns 4 (knights_tour):

Warnsdorff's heuristic rule

Raspuns 5 (csp_backtracking):

Complete assignment: {'X2': 3, 'X1': 1, 'X3': 2, 'X4': 1, 'X5': 2}

Raspuns 6 (minmax_alphabeta):

Root value: -2 (Alpha-Beta would visit fewer than 8 leaves)

Raspuns 7 (game_theory):

Pure Nash equilibria: (row=0, col=0) payoff=(5, 5), (row=1, col=0) payoff=(4, 3), (row=2, col=2) payoff=(4, 5).